Game Theory

Outlies

- Introduction to Game
- Key Concept in Game Theory

Game Theory

- **Game theory** is a branch of mathematics and economics that studies strategic interactions between rational decision-makers, known as players, in situations where the outcome of each player's decision depends on the decisions of others. It provides a framework for analyzing and predicting the behavior of individuals, firms, governments, and other entities in competitive or cooperative settings.

Players:

- **Players**: The individuals, firms, or entities making decisions in the game.

Strategies

Strategies: The possible actions or choices available to each player.

Payoffs

- **Payoffs**: The outcomes or rewards associated with different combinations of strategies chosen by the players.

Normal Form Games:

- **Normal Form Games**: Games represented by a matrix that shows the possible strategies of each player and the corresponding payoffs.

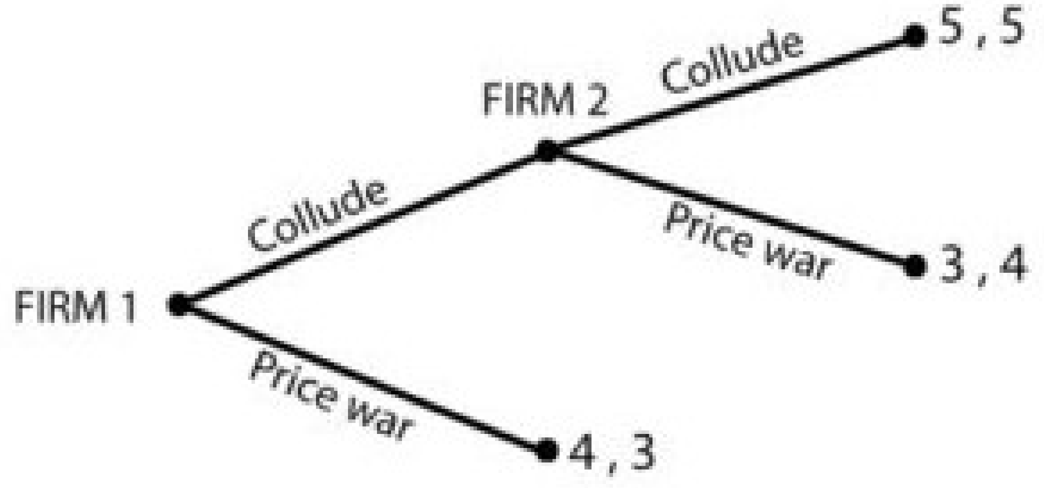
Example of Normal Form Game

		Firm 2	
		Develop Technology	Don't Develop Technology
Firm 1	Develop Technology	(1, 1)	(4, 0)
	Don't Develop Technology	(0, 4)	(2, 2)

Extensive Form Games:

- **Extensive Form Games**: Games represented by a tree-like structure that captures the sequence of moves and decisions made by the players.

Extensive Form Games



Dominant Strategy

- A dominant strategy in game theory refers to a strategy that yields the highest payoff for a player regardless of the strategies chosen by other players. In other words, a dominant strategy is the best course of action for a player, no matter what the opponents do.

Dominant Strategy

		Firm 2	
		High	Low
Firm 1	High	(15, 15)	(0, 25)
	Low	(25, 0)	(5, 5)

Pure strategy

- a pure strategy refers to a specific course of action or decision that a player chooses to follow with certainty. It is a strategy in which the player selects a single action from the available set of actions without any randomness or probability involved.

Example

- For example, consider a simple game between two players, Player A and Player B. Each player has two possible strategies: Strategy 1 or Strategy 2.
- If Player A decides to always choose Strategy 1, regardless of the actions taken by Player B, then Player A is employing a pure strategy. Similarly, if Player B decides to always choose Strategy 2, regardless of the actions taken by Player A, then Player B is also employing a pure strategy.

Mixed Strategy

- A mixed strategy in game theory is when a player makes decisions by randomly choosing between available options with specific probabilities.

Example

- Suppose there is a two-player game where each player, Player A and Player B, can choose between two actions: Action 1 or Action 2.
- Player A decides to adopt a mixed strategy, where they choose Action 1 with probability p and Action 2 with probability $1-p$, where $0 \le p \le 1$.
- Similarly, Player B also decides to adopt a mixed strategy, where they choose Action 1 with probability q and Action 2 with probability $1-q$, where $0 \le q \le 1$.

Nash Equilibrium

Nash Equilibrium is a concept in game theory where each player's strategy is optimal given the strategies chosen by the other players, with no player having an incentive to unilaterally deviate from their chosen strategy.

Prisoner's Dilemma

- In the Prisoner's Dilemma, two suspects are arrested and placed in separate cells. They are given the option to either "cooperate" by remaining silent or "defect" by confessing to the crime. The payoffs for each possible outcome are as follows:

Nash Equilibrium using the classic game called the "Prisoner's Dilemma."

		Column	
		Confess	Don't
Row	Confess	(–10, –10)	(0, –20)
	Don't	(–20, 0)	(–1, –1)

Pure strategy Nash Equilibrium using a payoff matrix.

- Suppose we have a two-player game where each player, Player A and Player B, can choose between two strategies: Strategy 1 or Strategy 2. Here's the payoff matrix for the game:

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                 | Strategy 1   | Strategy 2
------------------------------------------
Player A  |    3, 2        |    0, 1
Player B  |    2, 3        |    1, 0
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- The first number in each cell represents the payoff to Player A.
- The second number represents the payoff to Player B.
- Now, let's analyze the pure strategy Nash Equilibrium:
- If Player A chooses Strategy 1:
- Player B's best response is to choose Strategy 1, as it results in a higher payoff (3 vs. 2).
- Therefore, (Strategy 1, Strategy 1) is a Nash Equilibrium.
- If Player A chooses Strategy 2:
- Player B's best response is to choose Strategy 2, as it results in a higher payoff (1 vs. 0).
- Therefore, (Strategy 2, Strategy 2) is a Nash Equilibrium.
- In this case, there are two pure strategy Nash Equilibria:
- (Strategy 1, Strategy 1)
- (Strategy 2, Strategy 2)
- In both equilibria, each player's chosen strategy is optimal given the strategy chosen by the other player, with no player having an incentive to unilaterally deviate from their chosen strategy.

Bargaining

- In game theory, bargaining refers to the strategic interaction between two or more decision-makers (players) who seek to reach an agreement on the division of a set of resources or the outcome of a particular situation. Game theory provides a framework for analyzing bargaining situations, where players must consider the possible actions of others and their own incentives when making decisions.

Formation of beliefs

Formation of beliefs in game theory involves players developing expectations about the actions and strategies of others, influencing their own decisions; for instance, in repeated games like the Prisoner's Dilemma, players may update beliefs based on past actions to anticipate future behavior.

Formation of beliefs

- Formation of beliefs refers to the process through which individuals develop convictions or understandings about the world based on available information, personal experiences, and cognitive processes, shaping their perceptions and decision-making.
- Example: In financial markets, investors form beliefs about future stock prices based on factors such as company performance, economic indicators, and market trends, influencing their investment decisions.

Stackelberg's model of duopoly

- Stackelberg's model of duopoly is a game-theoretic framework where one firm, the leader, chooses its quantity, and the other firm, the follower, observes the leader's choice and responds strategically. For example, in the market for smartphones, if Apple (the leader) sets its price first, Samsung (the follower) observes Apple's price and adjusts its own pricing strategy accordingly.

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                      | Firm B HQ   | Firm B LQ
---------------------------------------------
Firm A HQ (Leader) | (5, 3)          | (6, 2)
Firm A LQ             | (4, 4)          | (3, 5)
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